Math 141, S06 **Name:** key

Quiz 11 **Section:** **SSN: xxx-xx-**

<u>Instructions:</u> There are totally 1 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Let $f(x) = x^3 - 2x - 2$. The equation $f(x) = 0$ has one real solution. Approximate it by **Newton's method**.

(a) Write down the iterative formula of Newton's method for solving $f(x) = 0$.

$f'(x) = 3x^2 - 2$

The iterative formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$
$$= x_n - \frac{x_n^3 - 2x_n - 2}{3x_n^2 - 2}$$

(b) Let the initial approximation $x_1 = 2$. Compute x_2, x_3 and x_4 by the above formula.

$$x_2 = x_1 - \frac{x_1^3 - 2x_1 - 2}{3x_1^2 - 2}$$
$$= 2 - \frac{2^3 - 2\cdot 2 - 2}{3\cdot 2^2 - 2}$$
$$= 2 - \frac{2}{10} = \frac{9}{5}$$
$$x_3 = \frac{9}{5} - \frac{(\frac{9}{5})^3 - 2\cdot(\frac{9}{5}) - 2}{3\cdot(\frac{9}{5})^2 - 2}$$
$$\approx 1.76994819$$

$x_4 \approx 1.76929266$